

Practice 1: Fraction meaning

Name: _____ Date: _____

For each problem, show your model or reasoning.

1. A rectangle is split into 8 equal parts. 3 are shaded. Write the fraction shaded.

Work space

2. In $\frac{5}{7}$, identify the numerator and denominator. Explain what each tells you.

Work space

3. Draw a bar model for $\frac{4}{6}$. Are the parts equal? Explain.

Work space

Practice 2: Fractions of sets

Name: _____ Date: _____

1. What is $\frac{3}{5}$ of 20 counters? Draw groups to show it.

Work space

2. 7 of 12 marbles are blue. Write the fraction that are blue and the fraction that are not blue.

Work space

3. A class has 24 students. $\frac{1}{4}$ bring a lunch from home. How many students is that?

Work space

Practice 3: Unit fractions

Name: _____ Date: _____

1. Write the unit fraction for one of 9 equal parts.

Work space

2. How many $\frac{1}{8}$ pieces make one whole? Explain with a strip.

Work space

3. Which is larger, $\frac{1}{3}$ or $\frac{1}{6}$? Use a drawing or words to justify.

Work space

Practice 4: Improper fractions and mixed numbers

Name: _____ Date: _____

1. Change $7\frac{3}{4}$ to a mixed number.

Work space

2. Change $2\frac{3}{5}$ to an improper fraction.

Work space

3. Draw a model for $1\frac{1}{2}$ and label the improper fraction.

Work space

Practice 5: Number lines

Name: _____ Date: _____

1. Divide 0 to 1 into 6 equal intervals and mark $\frac{4}{6}$.

Work space

2. Which fraction is at the third tick after 0 when a number line has eighths?

Work space

3. Mark $\frac{5}{4}$ on a number line. Write it as a mixed number.

Work space

Practice 6: Equivalent fractions by multiplying

Name: _____ Date: _____

1. Complete $\frac{2}{3} = \frac{?}{12}$.

Work space

2. Complete $\frac{5}{8} = \frac{15}{?}$.

Work space

3. Write two fractions equivalent to $\frac{3}{7}$.

Work space

Practice 7: Equivalent fractions by dividing

Name: _____ Date: _____

1. Simplify $12/18$.

Work space

2. Simplify $20/25$.

Work space

3. Is $18/24$ equivalent to $3/4$? Show how you know.

Work space

Practice 8: Visual equivalence

Name: _____ Date: _____

1. Explain why $\frac{1}{2}$ and $\frac{4}{8}$ name the same amount.

Work space

2. A strip has 3 of 6 equal parts shaded. Write two equivalent fractions.

Work space

3. True or false: $\frac{2}{5} = \frac{4}{10}$. Prove your answer with a model or calculation.

Work space

Practice 9: Missing numerators

Name: _____ Date: _____

1. $\frac{3}{4} = \frac{?}{20}$.

Work space

2. $\frac{7}{9} = \frac{21}{?}$.

Work space

3. $\frac{4}{11} = \frac{?}{33}$.

Work space

Practice 10: Missing denominators

Name: _____ Date: _____

1. $6/10 = 3/?$.

Work space

2. $8/12 = 2/?$.

Work space

3. $15/35 = 3/?$.

Work space

Practice 11: Simplest form

Name: _____ Date: _____

1. Reduce $16/20$ to simplest form.

Work space

2. Reduce $21/28$ to simplest form.

Work space

3. Is $9/13$ already in simplest form? Explain.

Work space

Practice 12: Equivalent fractions in context

Name: _____ Date: _____

1. Mia ate $\frac{2}{6}$ of a pizza. Ben ate $\frac{1}{3}$. Did they eat the same amount? Explain.

Work space

2. A recipe uses $\frac{3}{4}$ cup. How many eighths of a cup is that?

Work space

3. A trail is $\frac{6}{10}$ mile completed. Give the simplest equivalent fraction.

Work space

Practice 13: Compare using equivalence

Name: _____ Date: _____

1. Compare $\frac{2}{3}$ and $\frac{8}{12}$ using $<$, $>$, or $=$.

Work space

2. Compare $\frac{5}{6}$ and $\frac{15}{18}$. Explain.

Work space

3. Which is greater: $\frac{3}{5}$ or $\frac{9}{15}$? Show an equivalent fraction.

Work space

Practice 14: Fractions on models

Name: _____ Date: _____

1. Draw two equal-size rectangles to show $\frac{2}{3} = \frac{6}{9}$. Show your two rectangles clearly.

Work space

2. Shade a model showing $\frac{3}{4} = \frac{9}{12}$. Label the parts you shade.

Work space

3. Draw a set model showing $\frac{2}{5} = \frac{6}{15}$. Make each set part equal in size.

Work space

Practice 15: Explain and justify

Name: _____ Date: _____

1. Explain why multiplying both numerator and denominator by 3 keeps a fraction's value unchanged.

Work space

2. A student says $\frac{4}{8}$ is greater than $\frac{1}{2}$ because 4 is greater than 1. Explain the error.

Work space

3. Give a fraction equivalent to $\frac{5}{6}$ with denominator 24 and explain your steps.

Work space

Practice 16: Mixed and improper equivalence

Name: _____ Date: _____

1. Write $9/4$ as a mixed number and give an equivalent improper fraction for your mixed number.

Work space

2. Complete $1\frac{2}{3} = ?/3$.

Work space

3. Are $2\frac{1}{2}$ and $5/2$ equivalent? Show why.

Work space

Practice 17: Find the whole

Name: _____ Date: _____

1. If 4 parts represent $\frac{2}{3}$ of a collection, how many parts represent the whole? Explain.

Work space

2. 6 red tiles are $\frac{3}{5}$ of all the tiles. How many tiles are there?

Work space

3. A ribbon's $\frac{5}{8}$ is 15 cm. What is the full length?

Work space

Practice 18: Error analysis

Name: _____ Date: _____

1. Jordan changes $\frac{3}{5}$ to $\frac{6}{10}$ by adding 3 and 5. Is this reasoning valid? Correctly explain.

Work space

2. A student simplifies $\frac{12}{16}$ to $\frac{6}{8}$ and stops. Is it simplest form? Finish the work.

Work space

3. Find and correct the mistake: $\frac{2}{7} = \frac{2}{14}$.

Work space

Practice 19: Review

Name: _____ Date: _____

1. Simplify $\frac{24}{32}$, then write an equivalent fraction with denominator 20.

Work space

2. A garden has 18 of 30 plants flowering. Write the fraction, simplify it, and name an equivalent fraction with denominator 5.

Work space

3. Place $\frac{3}{2}$, $\frac{4}{4}$, and $\frac{3}{4}$ in order from least to greatest. Explain.

Work space

Practice 20: Final challenge

Name: _____ Date: _____

1. A 24-slice tray has 18 slices left. Write the fraction left in simplest form and an equivalent fraction with denominator 8.

Work space

2. Create three different fractions equivalent to $\frac{4}{5}$. Show the multiplier used each time.

Work space

3. Write a short rule for making equivalent fractions, then use it to show $\frac{7}{10} = \frac{?}{40}$.

Work space